Learning Reaction Mechanisms to Design Selective Catalysts

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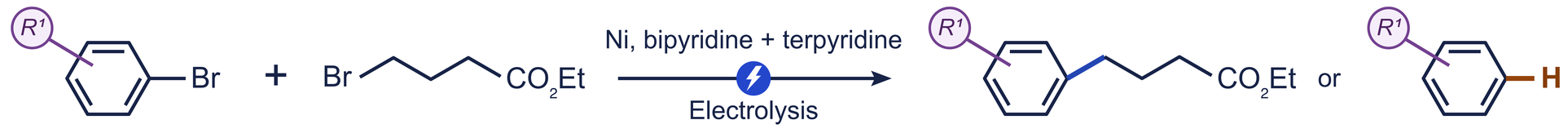
Meta AI Mentorship Program · 2024-2026

Rowland Fellowship finalist · 2021

When the substrate changes, what should change in the catalyst?

We'll explore how substrate, catalyst and reaction conditions control pathways and rates, then design catalysts and choose conditions for new substrates.

Proposed aryl–alkyl coupling



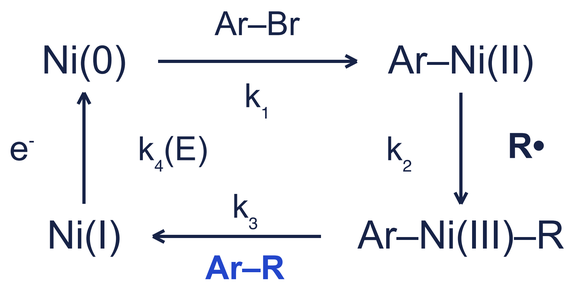
Aryl bromide series — **Fixed alkyl bromide** — **C–C coupling** — **Aryl reduction**

1. Predicting Pathways and Rates Across Substrates

Ar–Br + R–Br

Ligand + conditions

Mechanism model



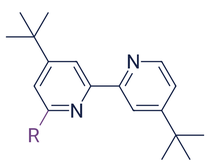
Candidate cycle and step rates

2. Designing Catalysts and Electrochemical Interfaces

Reactants Ar–Br + R–Br → **Desired product** Ar–R

Design model

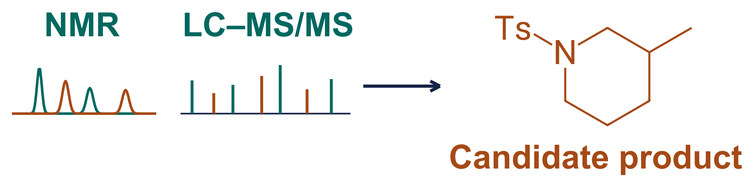
Pathways and rates guide design revisions.



Backbone + R + **M** (Metal) + Electrode, Solvent, Electrolyte

3. Uncovering Reaction Pathways through Untargeted Products

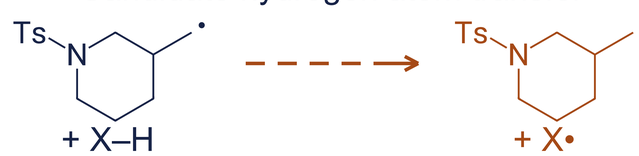
Spectral model



Candidate product

Pathway model

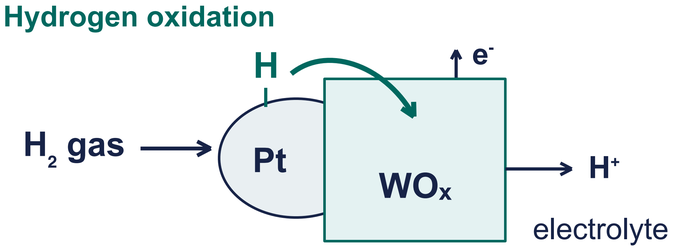
Candidate hydrogen atom transfer



Research foundations

Experiment

Hydrogen oxidation



Hydrogen transfer from platinum to the oxide primarily controlled the rate.

Connecting experiments and ML

How should we update our model with new result?

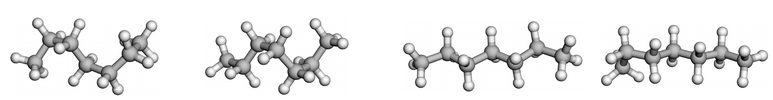
Rates, products and spectra

What does our model suggest we explore next?

Catalysts, conditions, experiments

Machine Learning

Solution — **Protein pocket**



Reference — Prediction — Reference — Prediction

EVA-Flow learns how molecules' conformation depends on environments

First-author publications

Experimental catalysis

1. Electrochemical Activation of C–C Bonds through Mediated Hydrogen Atom Transfer Reactions. **ChemSusChem 2022**
2. Mixed Electron–Proton Conductors Enable Spatial Separation of Bond Activation and Charge Transfer in Electrocatalysis. **JACS 2019**
3. Surface Restructuring of Nickel Sulfide Generates Optimally Coordinated Active Sites for Oxygen Reduction Catalysis. **Joule 2017**
4. A Membrane-Free Neutral pH Formate Fuel Cell Enabled by a Selective Nickel Sulfide Oxygen Reduction Catalyst. **Angew. Chem. Int. Ed. 2017**

Machine learning for chemistry

5. EVA-Flow: Learning Shared Chemistry for Unified Environment-Aware Molecular Conformation Generation. **TMLR 2026**
6. AskChem: Claim-Centered Infrastructure for Chemistry Literature Synthesis. **EMNLP System Demonstrations 2026**
7. ReasonFlow: Flow Matching that Thinks Step by Step. **Findings of EMNLP 2026**
8. Inconsistency of LLMs in Molecular Representations. **Digital Discovery 2025**
9. CatScore: Evaluating Asymmetric Catalyst Design at High Efficiency. **Digital Discovery 2024**